

Basil Varghesekutty

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PROFESSIONAL SUMMARY

AI-focused Computer Science graduate with hands-on experience building real-world machine learning and AI-powered applications. Developed RAG systems, computer vision solutions, and full-stack tools integrating LLM APIs and backend services. Comfortable working across Python, REST APIs, and modern AI tooling to turn ideas into functional products. Looking to contribute to fast-moving teams building scalable AI features and intelligent automation systems.

TECHNICAL SKILLS

- **AI/ML** PyTorch, TensorFlow, Scikit-learn, LangChain, Pandas, NumPy, ChromaDB, PowerBI, Keras.
- **Languages:** Python,C, SQL, JavaScript, TypeScript.
- **Frameworks:** Next.js, Node.js, Streamlit, Docker, Git/GitHub, AWS, Supabase, PostgreSQL, n8n.
- **Core Concepts:** Retrieval-Augmented Generation (RAG), Computer Vision, ReAct Framework, REST APIs, Natural Language Processing, ETL, AI Integration.

TECHNICAL TRAINING

SMECLabs

Data Science & AI Training Program

Kochi, India

Sep 2025 – Mar 2026

- Completed coursework in Python programming, machine learning algorithms (classification, regression), and data visualization.
- Cleaned, preprocessed, and analyzed raw datasets using Pandas and NumPy for capstone projects.
- Trained predictive models using Scikit-learn and built interactive data dashboards with Streamlit to display results.

KEY PROJECTS

Context-Aware Document Assistant

Python, LangChain, ChromaDB

- Built a RAG-based tool to parse, chunk, and query local PDF documents to retrieve specific technical information using LLM APIs.
- Applied the ReAct(Reason+Act) framework to allow the LLM agent to self-correct and perform multi-step reasoning during retrieval.
- Implemented semantic search using ChromaDB to store vector embeddings, reducing hallucinations by grounding answers in document context.

Real-Time Violence Detection System

3D CNN, TensorFlow, OpenCV

- Developed and trained a custom 3D Convolutional Neural Network using TensorFlow to extract spatiotemporal features for detecting physical altercations.
- Utilized OpenCV for real-time frame extraction and preprocessing, optimizing the pipeline for low-latency inference.
- Integrated the Twilio API to architect an automated alert system that triggers SMS notifications upon high-confidence event detection.

Attendance Monitoring Dashboard

Python, FaceNet, Next.js, Supabase

- Built a facial recognition pipeline using RetinaFace for face detection and FaceNet for generating embeddings and user identification.
- Developed a full-stack administrative dashboard using Next.js and Supabase, integrating real-time attendance tracking with secure user authentication and persistent cloud storage.

AI Chatbot

Python, LangChain, Streamlit

- Built a modular Python chat engine integrating Groq LLMs LangChain with dotenv-based API key management and configurable model selection.
- Implemented sliding-window conversation memory and prompt orchestration for context-aware replies.

EDUCATION

Government Engineering College Idukki – B.Tech Computer Science 2021 – 2025 *Relevant Coursework: Data Structures Algorithms, Machine Learning, Deep Learning, Artificial Intelligence,Data Mining, DBMS.*

AWARDS CERTIFICATIONS

- **Project Scholarship – IITM Pravartak Technologies:** Awarded a one-year funding grant to support technical project development. Jan 2025 – Jan 2026
- **Professional Certification in Data Science with AI ML** 2025
- **NPTEL Certification on Machine Learning (Silver Elite)** 2024

LEADERSHIP EXTRACURRICULARS

- **Technical Coordinator, IEEE PES (2022–23):** Organized and coordinated technical events and student hackathons.
- **Arts Club Secretary (2023–25):** Managed event logistics, scheduling, and team operations for major college functions.